sent off-chip as an analogue signal. All of the pixels can be devoted to light capture and the output's uniformity, which is a key factor in image quality, is high.

On the other hand, in a CMOS sensor, each pixel has its own charge-to-voltage convertor, amplifier and a pixel-select switch (Fig. 7). This is called active-pixel sensor architecture in contrast to passive-pixel sensor architecture used in a CCD sensor. Also, the sensor often includes on-chip amplifiers, noise-correction and analogue-to-digital conversion circuits, and other circuits critical to pixel sensors' operation. The chip in this case outputs digital bits. Inclusion of these functions reduces the area available for light capture. Also, with each pixel doing its own conversion, uniformity and consequently image quality is lower. While readout mechanism of a CCD sensor is serial, it is massively parallel in the case of a CMOS sensor, allowing high total bandwidth for high speed.

CCD sensors versus CMOS sensors

Some of the key differences between CCD and CMOS sensors include:

1. Fabrication of CMOS sensors involves standard CMOS technology well established for fabrication of integrated circuits. This also allows easy integration of required electronics on the same chip, thereby resulting in devices that are compact and cost-effective. On the other hand, CCD sensor fabrication involves dedicated and costly manufacturing processes.

2. Compared to CCD sensors, CMOS sensors have relatively poor sensitivity and uniformity. The poor sensitivity is due to poor fill factor, while poor uniformity is due to the use of separate amplifiers for different pixels as against a single amplifier for all pixels in the case of CCD sensors.

3. CMOS sensors have higher speed than CCD devices due to the use of active pixels and inclusion of analogue-to-digital converters on the same chip leading to smaller propagation delays. Low-end CMOS sensors have low power requirements, but high-speed CMOS cameras typically require more power than CCD devices.

4. CCD sensors have higher dynamic range than CMOS sensors.

5. When it comes to temporal noise, CMOS sensors score over CCD sensors due to lower bandwidth of amplifiers at each pixel. But in terms of fixed-pattern noise performance, CCD sensors are better due to single-point charge-to-voltage conversion.

6. CMOS sensors allow on-chip incorporation of a range of functions such as automatic gain control, auto exposure control, image compression, colour encoding and motion tracking.

7. Due to the absence of shift registers, CMOS sensors are immune to smearing around over-exposed pixels, which is caused by spilling of charge into the shift register.

CCD sensors are used in cameras that offer excellent photo sensitivity and focus on high-quality, high-resolution images. CMOS sensors, on the other hand, are generally characterised by lower image quality, resolution and photo sensitivity. These are usually less expensive and have longer battery life due to lower power consumption. Also, these are fast achieving near parity with CCD devices in some applications.

Ladar sensors

Laser radars, also called Ladars, use a laser beam instead of microwaves. That is, in laser radars the transmitted electromagnetic energy lies in the optical spectrum, whereas in microwave radars it is in the microwave region. Frequencies associated with laser radars are very high, ranging from 30THz to 300THz. The corresponding wavelengths are 10μm to 1.0μm. Higher operating frequency means higher operating bandwidth, greater time or range resolution, and enhanced angular resolution. Another advantage of laser radars over microwave radars is their immunity to jamming.

Higher frequencies associated with laser radars permit detection of smaller objects. This is made possible by the fact that laser radar output wavelengths are much smaller than the smallest-sized practical objects. In other words, laser radar cross-